

4.6

$$a. Pr(Y=1 | X_1=40, X_2=3.5) = \frac{1}{1 + \exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2)} = \frac{1}{1 + \exp(-110.05 \times 40 + 1 \times 3.5)} = \frac{1}{1 + \exp(1.5)} = 0.3775$$

$$b. G(\beta_0, \beta_1, \beta_2) = \frac{1}{1 + e^{-t}} \Rightarrow 0.5(1 + e^{-t}) = 1 \Rightarrow t = 0$$

$$Y = \beta_0 + \beta_1 X_1$$

$$\Rightarrow 0 = -6 + 0.05 X_1 + 3.5 \Rightarrow X_1 = 50$$

4.9

$$a. odds = \frac{P}{1-P} = 0.37$$

$$\Rightarrow P = 0.27$$

$$b. odds = \frac{0.16}{1-0.16} = 0.19$$

$$4.12 \quad a. \log odds = \log \left(\frac{\hat{Pr}(Y=Orange | X=x)}{\hat{Pr}(Y=Apple | X=x)} \right) = \log \left(\frac{\frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 x)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 x)}}{\frac{1}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 x)}} \right) = \log(\exp(\hat{\beta}_0 + \hat{\beta}_1 x)) = \hat{\beta}_0 + \hat{\beta}_1 x$$

$$b. \log odds = \log \left(\frac{\hat{Pr}(Y=Orange | X=x)}{\hat{Pr}(Y=Apple | X=x)} \right) = \log \left(\frac{\frac{\exp(\hat{\alpha}_{orange} + \hat{\alpha}_{orange,1} x)}{\exp(\hat{\alpha}_{orange} + \hat{\alpha}_{orange,1} x) + \exp(\hat{\alpha}_{apple} + \hat{\alpha}_{apple,1} x)}}{\frac{\exp(\hat{\alpha}_{apple} + \hat{\alpha}_{apple,1} x)}{\exp(\hat{\alpha}_{orange} + \hat{\alpha}_{orange,1} x) + \exp(\hat{\alpha}_{apple} + \hat{\alpha}_{apple,1} x)}} \right)$$

$$= (\hat{\alpha}_{orange} - \hat{\alpha}_{apple}) + (\hat{\alpha}_{orange,1} - \hat{\alpha}_{apple,1}) x$$

$$c. \hat{\beta}_0 + \hat{\beta}_1 x = (\hat{\alpha}_{orange} - \hat{\alpha}_{apple}) + (\hat{\alpha}_{orange,1} - \hat{\alpha}_{apple,1}) x = 2 - x \Rightarrow ((\hat{\alpha}_{orange} - \hat{\alpha}_{apple}), (\hat{\alpha}_{orange,1} - \hat{\alpha}_{apple,1})) = [2, -1]$$

$$d. \hat{\beta}_0 + \hat{\beta}_1 x = (\hat{\alpha}_{orange} - \hat{\alpha}_{apple}) + (\hat{\alpha}_{orange,1} - \hat{\alpha}_{apple,1}) x = (1.2-3) + (-2-0.6)x \Rightarrow [\hat{\beta}_0, \hat{\beta}_1] = [-1.8, -2.6]$$

e. The predicted class labels of mine and my friend's model will agree 100% of the time as their log odds are the same.